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Some properties of k -Delaunay and k -Gabriel graphs

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ABSTRACT

We consider two classes of higher order proximity graphs defined on a set of points in the plane, namely, the k -Delaunay graph and the k -Gabriel graph. We give bounds on the following combinatorial and geometric properties of these graphs: spanning ratio, diameter, connectivity, chromatic number, and minimum number of layers necessary to partition the edges of the graphs so that no two edges of the same layer cross.

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1. Introduction

Let S be a set of n points in the plane in general position (no three are collinear and no four are concyclic). A proximity graph on S is a geometric graph where two points are adjacent if they satisfy some specific proximity criterion. Proximity graphs have been widely studied due to their theoretical interest and to their applications in situations where it is necessary to extract the “shape” of a set of points (see [16] for a survey).

Adjacency in many proximity graphs is defined in terms of an empty region associated to any pair of points. To provide more flexibility the definition of the graphs can be relaxed to allow up to k points to lie in the neighborhood region. This gives rise to higher order proximity graphs. In this paper we deal with two such graphs.

We consider the k -Delaunay graph of S (denoted k -DG(S)), where a straight-line segment connects points $p_i, p_j \in S$ if there exists a circle $C(p_i, p_j)$ through p_i and p_j with at most k points of S in its interior. The standard Delaunay triangulation corresponds to 0-DG(S) and will be denoted by DT(S).

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Table 1

Worst-case possible values for the studied properties. SR , D , κ , χ , and θ_c denote the spanning ratio, diameter, connectivity, chromatic number, and constrained geometric thickness, respectively. The results hold for specific ranges of k ; see the complete statements throughout the paper.

Gabriel graphs	Delaunay graphs
$\Theta(\sqrt{\frac{n}{k}}) \leq \max_{ S =n} SR(k\text{-GG}(S)) \in O(\sqrt{n})$	$1.5931 \leq \max_{ S =n} SR(k\text{-DG}(S)) \leq 1.998$
$\lceil \frac{n-1}{k+1} \rceil \leq \max_{ S =n} D(k\text{-GG}(S)) \leq \frac{n-2}{k+1} + 1$	$\lfloor \frac{n}{2(k+1)} \rfloor \leq \max_{ S =n} D(k\text{-DG}(S)) \leq \frac{n-2}{2(k+1)} + 1$
$\min_{ S =n} \kappa(k\text{-GG}(S)) = k + 1$	$\min_{ S =n} \kappa(k\text{-DG}(S)) = 2(k + 1)$
$3(k + 1) \leq \max_{ S =n} \chi(k\text{-GG}(S)) \leq 6(k + 1)$	$4(k + 1) \leq \max_{ S =n} \chi(k\text{-DG}(S)) \leq 6(k + 1)$
$3 \leq \max_{ S =n} \theta_c(1\text{-GG}(S)) \leq 4$	$3 \leq \max_{ S =n} \theta_c(1\text{-DG}(S)) \leq 4$
$\frac{3k}{2} \leq \max_{ S =n} \theta_c(k\text{-GG}(S)) \leq 18k^2 + 3k$	$\frac{3k}{2} \leq \max_{ S =n} \theta_c(k\text{-DG}(S)) \leq 18k^2 + 3k$

We also study the k -Gabriel graph of S (denoted by $k\text{-GG}(S)$), where a straight-line segment connects points $p_i, p_j \in S$ if the closed disk centered at the midpoint of the segment $\overline{p_i p_j}$ with both p_i and p_j on its boundary contains at most k points from S different from p_i, p_j . The standard Gabriel graph corresponds to $0\text{-GG}(S)$ and will be denoted by $\text{GG}(S)$.

The combinatorial and geometric properties of these graphs have been widely studied for the case $k = 0$ (see [16]). However, not so much is known for higher values of k . In [22] it has been shown that $k\text{-DG}(S)$ and $k\text{-GG}(S)$ can be computed in $O(k^2 n \log n)$ time, and that the k -Gabriel graph can be used to obtain fast algorithms for the Euclidean bottleneck biconnected edge subgraph problem and for the Euclidean bottleneck matching problem. The structure of the set of triangulations that can be constructed with edges of the k -Delaunay graph is analyzed in [1], as well as the minimum and maximum number of edges of this graph, and a use of $k\text{-DG}(S)$ in a coloring problem with applications to cellular networks. Finally, the number of crossings in these graphs is discussed in [3].

In this paper we investigate other properties of these graphs. The first property we consider is the spanning ratio, a parameter capturing to what extent traveling along a graph is much longer compared to traveling along the plane (the formal definition is given below). Our goal is to study the relationship between k and the spanning ratio. For $k = 0$, the spanning ratio of several proximity graphs has been studied in the literature. In particular, it has been shown the spanning ratio of the Gabriel graph might be as bad as $\Theta(\sqrt{n})$ [6]. On the other hand, it is known that the spanning ratio of the Delaunay triangulation of any set of points in the plane is constant, but determining the exact value of the spanning ratio of this graph remains a challenging open problem [7,11,17,24,25].

We also study three graph theoretical properties of $k\text{-DG}(S)$ and $k\text{-GG}(S)$. These properties are the diameter, which can be seen as a combinatorial counterpart to the spanning ratio, the connectivity and the chromatic number. In the three cases the bounds we provide are tight (up to multiplicative constants).

Finally, we deal with a variation of the notion of *geometric thickness*. The geometric thickness of a graph G is the minimum number of layers such that there exists a straight-line drawing of G in the plane and a partition of the edges of G into layers such that no two edges of the same layer cross. The study of the geometric thickness finds applications in problems such as representing nonplanar graphs so that their complexity is minimized, or designing printed circuit boards, where connections that cross must lie in distinct layers [10]. In some cases one might not have the freedom of choosing the drawing of the graph because it is fixed or forced to satisfy additional restrictions. This motivates the use of a distinct parameter defined as follows. The *constrained geometric thickness* of a geometric graph G (i.e., a fixed straight-line drawing of a graph in the plane) is the smallest number of layers necessary to partition the edges of G into layers in such a way that no two edges of the same layer cross. The concept of constrained geometric thickness has appeared before in the literature, and in particular it is known that it can be tested in $O(n \log n)$ time whether the constrained geometric thickness of a geometric graph is 2 [12]. In this reference the constrained geometric thickness of a geometric graph is called “thickness of a drawing of a graph”, but in the current paper we prefer to use our notation because we see the proximity graphs not as abstract graphs, but as geometric graphs.

We give bounds on the constrained geometric thickness of the graphs $k\text{-DG}(S)$ and $k\text{-GG}(S)$. Notice that the algorithmic question of testing whether the constrained geometric thickness of a geometric graph is at most some given value k , for $k > 2$, is NP-complete [12]. The fact that, when it comes to the constrained geometric thickness, the embedding of the graph is fixed appears to be a natural restriction when dealing with these graphs, since in this case the position of the vertices and the adjacencies have a geometric meaning that is not preserved if vertices are allowed to move around.

From a theoretical point of view, determining the constrained geometric thickness of $k\text{-DG}(S)$ and $k\text{-GG}(S)$ is a particular case of a more general problem that remains unsolved (see, for example, [5,14,18]): for every geometric graph G with at most λ pairwise crossing edges, can the edges of G be colored with $f(\lambda)$ colors such that crossing edges receive distinct colors? In the case of $k\text{-GG}(S)$ and $k\text{-DG}(S)$ the answer is affirmative, as we show that these graphs contain at most $2k + 1$ pairwise crossing edges and we give a quadratic upper bound on the number of colors required.

From a more practical point of view, $\text{DT}(S)$ and $\text{GG}(S)$ satisfy some properties that make them interesting in the context of routing in wireless networks [8,19]. Finding ways to extract plane layers from $k\text{-DG}(S)$ or $k\text{-GG}(S)$ may have applications in this setting.

Our results are summarized in Table 1.

For all $k \geq 0$, the following relations hold: (i) $k\text{-DG}(S) \subseteq (k + 1)\text{-DG}(S)$, (ii) $k\text{-GG}(S) \subseteq (k + 1)\text{-GG}(S)$, (iii) $k\text{-GG}(S) \subseteq k\text{-DG}(S)$.

2. Spanning ratio

Let G be a geometric graph on S and $P = \{p_1 p_2 \cdots p_l\}$ be a path in G . We define the *geometric length* of P as $\sum_{i=1}^{l-1} |p_i p_{i+1}|$, where $|p_i p_j|$ is the Euclidean distance between p_i and p_j . The *geometric distance* between points $p_i, p_j \in S$, denoted by $d_g(p_i, p_j)$, is the minimum over the geometric length of all paths in G connecting p_i and p_j . The *spanning ratio* of G is defined as

$$SR(G) = \max_{p_i \neq p_j \in S} \frac{d_g(p_i, p_j)}{|p_i p_j|}.$$

The spanning ratio of $DT(S)$ is not greater than 1.998 [24]. Since k -DG(S) is a supergraph of $DT(S)$, we have:

Corollary 1. For any set S of n points in the plane, $SR(k$ -DG(S)) $\leq$ 1.998.

The number of edges of k -Delaunay graphs grows with k . Consequently, it would be reasonable to believe that the spanning ratio of these graphs decreases as k increases, and thus Corollary 1 can be refined. The next theorem shows somehow the opposite, i.e., that in the worst case the spanning ratio of k -DG is not smaller than the spanning ratio of the Delaunay triangulation.

Theorem 1. For any set S of n points in the plane, any $k \geq 1$, any $n' \geq (k+1)n$, and any $\varepsilon > 0$, there exists a set S' of n' points such that $SR(k$ -DG(S')) $\geq SR(DT(S)) - \varepsilon$.

Proof. Consider the Delaunay triangulation of S . Since S is in general position, the combinatorial structure of the graph does not change when moving each point in S at most ε' , for sufficiently small values of $\varepsilon' > 0$. The supremum of the values of ε' satisfying this property is called the *tolerance* of $DT(S)$ and is denoted by $\text{tol}(DT(S))$ [2].

Let i, j be a pair of indices such that

$$SR(DT(S)) = \frac{d_g(p_i, p_j)}{|p_i p_j|}.$$

Given $\varepsilon > 0$, for each $p_l \in S$, define $p_{l,0} = p_l$ and place k new points $p_{l,1}, p_{l,2}, \dots, p_{l,k}$ at distance from $p_{l,0}$ less than $\varepsilon' = \min\{\text{tol}(DT(S)), \frac{|p_i p_j| \varepsilon}{2(k+1)n}\}$ and in general position. If $n' > (k+1)n$, add extra points far from the set S . Let S' be the resulting set of points. In the following lines we prove that $SR(k$ -DG(S')) is almost as large as $SR(DT(S))$.

Suppose that p_i and p_m are not adjacent in $DT(S)$, and let v, ι be two integers such that $0 \leq v, \iota \leq k$. We next argue that, in this case, $p_{i,v}$ and $p_{m,\iota}$ are not adjacent in k -DG(S'). Consider a circle C through $p_{i,v}$ and $p_{m,\iota}$. We want to see that C contains at least $k+1$ points of S' in its interior. In the Delaunay triangulation of the set of points formed by $p_{i,v}$ and $p_{m,\iota}$, together with $\{p_{r,0}\}_{r \neq i,m}$, the points $p_{i,v}$ and $p_{m,\iota}$ are not connected, since this triangulation has the same combinatorial structure as $DT(S)$ by the choice of ε' . Hence, C contains at least one point of the group $\{p_{r,0}\}_{r \neq i,m}$. Analogously, C contains at least one point of the group $\{p_{r,\tau}\}_{r \neq i,m}$ for all $\tau \in \{1, \dots, k\}$. Consequently, C contains at least $k+1$ points of S' in its interior.

Let $p_{i,0}, p_{j,0} = p_i, p_j$, and $p_{i,0}, p_{i,1}, p_{i,2}, \dots, p_{i,h}, p_{j,0}$ be the shortest path from $p_{i,0}$ to $p_{j,0}$ in k -DG(S'). By the triangle inequality, $|p_{i,r}, p_{i,r+1}, p_{i,r+2}, \dots, p_{i,h}, p_{j,0}| \geq |p_{i,r}, p_{i,r+1}| - 2\varepsilon'$, for any $r \leq h-1$. Therefore, in k -DG(S'),

$$\begin{aligned} d_g(p_{i,0}, p_{j,0}) &= |p_{i,0}, p_{i,1}| + |p_{i,1}, p_{i,2}| + \cdots + |p_{i,h-1}, p_{i,h}| + |p_{i,h}, p_{j,0}| \\ &\geq |p_{i,0}, p_{i,1}| + |p_{i,1}, p_{i,2}| + \cdots + |p_{i,h-1}, p_{i,h}| - 2\varepsilon'h. \end{aligned}$$

Notice that $h \leq (k+1)n$. Let $r \in \{0, 1, \dots, h-1\}$. Since $p_{i,r}, p_{i,r+1}, p_{i,r+2}, \dots, p_{i,h}, p_{j,0}$ is an edge of k -DG(S'), we have that either $p_{i,r} = p_{i,r+1}$ or $p_{i,r}, p_{i,r+1}$ is an edge of $DT(S)$. Therefore $p_{i,0}, p_{i,1}, p_{i,2}, \dots, p_{i,h-1}, p_{i,h}$ is a walk in $DT(S)$ connecting p_i to p_j . Thus $|p_{i,0}, p_{i,1}| + |p_{i,1}, p_{i,2}| + \cdots + |p_{i,h-1}, p_{i,h}| \geq d_g(p_i, p_j)$. Combining these facts,

$$d_g(p_{i,0}, p_{j,0}) \geq d_g(p_i, p_j) - 2\varepsilon'(k+1)n \geq d_g(p_i, p_j) - |p_i p_j| \varepsilon.$$

Finally,

$$SR(k\text{-DG}(S')) \geq \frac{d_g(p_{i,0}, p_{j,0})}{|p_{i,0}, p_{j,0}|} \geq SR(DT(S)) - \varepsilon. \quad \square$$

Corollary 2. For any $k \geq 1$ and any sufficiently large value of n , there exist sets of n points in the plane whose k -Delaunay graphs have spanning ratio larger than 1.5931.

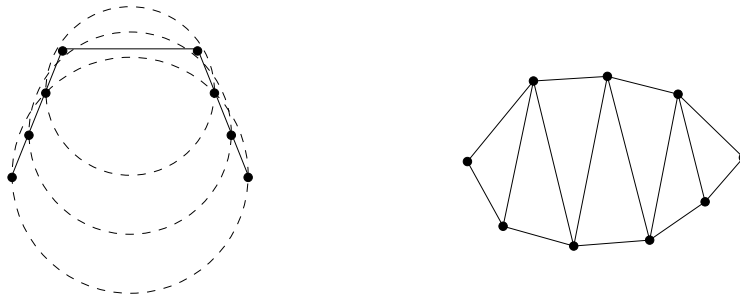


Fig. 1. Left: Gabriel graph tower construction of 8 points. Right: A sequential triangulation.

Proof. It suffices to consider the Delaunay triangulation described in [25] having spanning ratio at least 1.5932. Then we apply Theorem 1 with $\varepsilon = 10^{-4}$. $\square$

If $k \geq n/2 - 1$, the graph $k\text{-DG}(S)$ is the complete graph, so it has spanning ratio 1. If k is just one unit smaller, the spanning ratio of $k\text{-DG}(S)$ might be almost $\sqrt{2}$:

Remark 1. For any $n \geq 5$ and $\varepsilon > 0$, there exist sets of n points in the plane whose $(\lfloor \frac{n}{2} \rfloor - 2)$ -Delaunay graphs have spanning ratio larger than $\sqrt{2} - \varepsilon$.

Such a point set S can be constructed by placing two points p_1 and p_2 at distance 2, and two groups of $\lfloor n/2 \rfloor - 1$ and $\lfloor n/2 \rfloor - 1$ points at distinct sides of the segment $\overline{p_1 p_2}$ such that all points q in these groups are at distance approximately $\sqrt{2}$ from both p_1 and p_2 , and satisfy $\angle p_1 q p_2 > \pi/2$. The points p_1 and p_2 are not adjacent in $(\lfloor \frac{n}{2} \rfloor - 2)\text{-DG}(S)$.

Let us finally consider k -Gabriel graphs. The situation in this case is different because for $k = o(n)$ these graphs do not have constant spanning ratio:

Theorem 2. For any set S of n points in the plane and $k \leq n - 2$, the spanning ratio of $k\text{-GG}(S)$ is $O(\sqrt{n})$. There exist sets of n points in the plane whose k -Gabriel graphs have spanning ratio $\Theta(\sqrt{\frac{n}{k}})$.

Proof. The first part follows from a result in [6] stating that the spanning ratio of the 0-Gabriel graph of any n -point set is at most $\frac{4\pi}{3}\sqrt{2n} - 4$.

As for the second part, consider the Gabriel graph tower construction in [6] with $\lceil \frac{n}{k+1} \rceil$ points, which has spanning ratio $\Theta(\sqrt{\frac{n}{k}})$. (See Fig. 1, left.) Notice that all non-Gabriel edges are precluded by points lying in the interior of the closed disks associated to these edges. Consequently, for sufficiently small values of $\varepsilon' > 0$, each point can be moved at most ε' without changing the combinatorial structure of the graph. Now, proceeding as in the proof of Theorem 1, we obtain a point set whose k -Gabriel graph has spanning ratio $\Theta(\sqrt{\frac{n}{k}})$. $\square$

3. Diameter and connectivity

We define the *combinatorial length* of a path P on a geometric graph G as the number of its edges. The *combinatorial distance* between points $p_i, p_j \in S$, denoted by $d_c(p_i, p_j)$, is the minimum over the combinatorial length of all paths in G connecting p_i and p_j . The *diameter* of G , denoted by $D(G)$, is defined as the maximum over the combinatorial distance of all pairs of points in S .

The *connectivity* $\kappa(G)$ of a graph $G = (V, E)$ is the size of the smallest subset $V' \subseteq V$ such that the induced subgraph $G[V \setminus V']$ is disconnected.

In this section we determine the diameter and the connectivity of the k -Delaunay graph and the k -Gabriel graph. For any graph, these parameters are related in the following way:

Lemma 1. (See [23].) If G is a graph on n vertices with connectivity $\kappa(G) \geq 1$ and diameter $D(G) \geq 1$, then $n \geq \kappa(G)(D(G) - 1) - 2$.

We derive lower bounds for the connectivity of $k\text{-DG}(S)$ and $k\text{-GG}(S)$. Combining these bounds with Lemma 1, we obtain upper bounds for the diameter of these graphs. We start with k -Delaunay graphs.

Theorem 3. For any set S of n points in the plane and $k \leq \lceil n/2 \rceil - 2$, $\kappa(k\text{-DG}(S)) \geq 2(k + 1)$. There exist sets of n points in the plane whose k -Delaunay graphs have connectivity $2(k + 1)$.

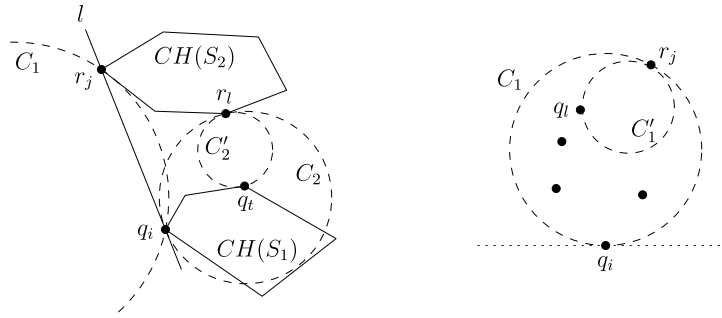


Fig. 2. The two cases in the proof of Theorem 3.

Proof. Let S' be a minimal subset of S such that $k\text{-DG}(S) \setminus S'$ is disconnected. Let $S_1, \dots, S_m$ be the vertex sets of the connected components of $k\text{-DG}(S) \setminus S'$. If $m = 1$, then $k\text{-DG}(S)$ is the complete graph and $\kappa(k\text{-DG}(S)) = n - 1 \geq 2(k + 1)$, for $k \leq \lceil n/2 \rceil - 2$. Otherwise we distinguish two cases.

We first consider the case where $m = 2$ and the two components can be separated by a horizontal line. We denote by $q_1, q_2, \dots$ the vertices in S_1 and by $r_1, r_2, \dots$ the vertices in S_2 . Suppose that the common tangent l of $CH(S_1)$ and $CH(S_2)$ leaving these polygons to its right passes through q_i and r_j . (See Fig. 2, left.) Then there exists a circle C_1 through q_i and r_j that does not contain any point from S to the right of the segment $\overline{q_i r_j}$. Since q_i and r_j belong to distinct components of $k\text{-DG}(S) \setminus S'$, (q_i, r_j) is not an edge of either $k\text{-DG}(S) \setminus S'$ or $k\text{-DG}(S)$. Therefore every circle through q_i and r_j contains at least $k + 1$ points of S in its interior. In particular, C_1 contains $k + 1$ such points, none of which can be in $S \setminus S'$ by construction. Hence C_1 contains $k + 1$ points of S' , and all of these points are to the left of l . If $S_1 = \{q_i\}$ and $S_2 = \{r_j\}$, we can construct a circle C_2 through q_i and r_j that does not contain any point from S to the left of the segment $\overline{q_i r_j}$. This circle contains at least $k + 1$ points of S' , and these points are different from the ones in C_1 because they are to the right of l . Thus $|S'| \geq 2(k + 1)$. Otherwise we can assume that $|S_2| \geq 2$, and we consider a circle C_2 tangent to l at q_i growing towards the right of l until it contains a point r_l on its boundary. If C_2 does not contain any point from S_1 in its interior, we define $C'_2 = C_2$. Otherwise, let C'_2 be the largest circle tangent to C_2 at r_l not containing any point from S_1 in its interior. Notice that C'_2 does not contain any point from S_2 in its interior and contains some point q_t on its boundary. Since r_l and q_t belong to distinct components of $k\text{-DG}(S) \setminus S'$, C'_2 contains at least $k + 1$ points of S' . These points are distinct from the points in C_1 because they are to the right of l , which yields that $|S'| \geq 2(k + 1)$.

In the second case there exists a vertex $q_i \in S \setminus S'$ such that each of the two open half-spaces separated by the horizontal line through q_i contains at least one point in $S \setminus S'$ belonging to a connected component distinct from q_i 's. Without loss of generality, we assume $q_i \in S_1$. Let C_1 be the smallest circle that has q_i at the south pole and includes a point r_j of a distinct connected component of $k\text{-DG}(S) \setminus S'$. (See Fig. 2, right.) By definition, points in C_1 belong to U_1 or S' . We shrink C_1 keeping the tangency on r_j until it contains exactly one point $q_l \in U_1$. Let C'_1 be the circle we obtain. Since the vertices r_j and q_l belong to distinct components, C'_1 contains at least $k + 1$ points from S' . By repeating the same idea with circles that have q_i as the north pole we obtain a circle C'_2 containing at least $k + 1$ points from S' , and whose interior is disjoint from the interior of C'_1 . Therefore, $|S'| \geq 2(k + 1)$.

It remains to prove the second part of the claim. We consider a set of $\lceil \frac{n}{k+1} \rceil$ points whose Delaunay triangulation is a sequential triangulation. A sequential triangulation can be constructed iteratively as follows: p_1, p_2 , and p_3 form a triangle and, for $i \geq 4$, p_i is adjacent to p_{i-1} and p_{i-2} (see Fig. 1, right for an example). These triangulations can be disconnected by removing two points. As in Theorem 1, each point (except possibly one) can be replaced by $k + 1$ points so that, in the k -Delaunay graph of the new point set, any two points can only be adjacent if they belong to the same cluster or their original points were adjacent. The k -Delaunay graph of this point set has connectivity at most $2(k + 1)$, because it can be disconnected by removing two clusters of $k + 1$ points corresponding to two points whose removal disconnects the original sequential triangulation. By the first part of the theorem, the connectivity of this k -Delaunay graph is exactly $2(k + 1)$. $\square$

Corollary 3. Let S be a set of n points in the plane and $k \leq \lceil n/2 \rceil - 2$. Then $D(k\text{-DG}(S)) \leq \frac{n-2}{2(k+1)} + 1$. There exist sets of n points in the plane whose k -Delaunay graphs have diameter greater than or equal to $\lfloor \frac{n}{2(k+1)} \rfloor$.

Proof. The upper bound follows from Theorem 3 and Lemma 1. To illustrate that it is tight (up to additive constants) we can use the same point set as in Theorem 3, whose k -Delaunay graph has diameter $\lfloor \frac{1}{2} \lceil \frac{n}{k+1} \rceil \rfloor$. $\square$

In general, the k -Gabriel graph has fewer edges than the k -Delaunay graph, so its connectivity is usually smaller:

Theorem 4. For any set S of n points in the plane and $k \leq n - 2$, $\kappa(k\text{-GG}(S)) \geq k + 1$. There exist sets of n points in the plane whose k -Gabriel graphs have connectivity $k + 1$.

Proof. Let S' be a minimal subset of S with the property that $k\text{-GG}(S) \setminus S'$ is disconnected. If $|S'| = n - 1$, then $k\text{-GG}(S)$ is the complete graph and $\kappa(k\text{-GG}(S)) = n - 1 \geq k + 1$, for $k \leq n - 2$. Otherwise, $k\text{-GG}(S) \setminus S'$ contains at least two connected components; let p_i and p_j be the pair of vertices at disjoint components of $k\text{-GG}(S) \setminus S'$ such that their distance in the plane is minimal. That is, any other pair of vertices $p_l, p_m \in S \setminus S'$ such that $|p_l p_m| < |p_i p_j|$ are in the same connected component of $k\text{-GG}(S) \setminus S'$.

Since p_i and p_j belong to distinct components of $k\text{-GG}(S) \setminus S'$, the edge (p_i, p_j) is not present in $k\text{-GG}(S)$. Therefore, the closed disk centered at the midpoint of the segment $\overline{p_i p_j}$ with both p_i and p_j on its boundary contains at least $k + 1$ points from S different from p_i, p_j . Let p_l be one of these points; notice that p_l does not belong to $S \setminus S'$: otherwise we would have a contradiction with the definition of p_i and p_j (since both $|p_l p_i| < |p_i p_j|$ and $|p_l p_j| < |p_i p_j|$ hold, and p_l cannot belong to both the components of p_i and p_j). Thus, $|S'| \geq k + 1$.

Finally, we describe a point set showing that this bound is tight. Let $S = \{p_1, \dots, p_n\}$ be a set of n points sorted by x coordinate in an infinitesimally perturbed horizontal line. Then $k\text{-GG}(S)$ contains the edge (p_i, p_j) if and only if $|i - j| \leq k + 1$. Thus removing $p_2, p_3, \dots, p_{k+2}$ disconnects p_1 from the rest of the graph. $\square$

Corollary 4. For any set S of n points in the plane and $k \leq n - 2$, $D(k\text{-GG}(S)) \leq \frac{n-2}{k+1} + 1$. There exist sets of n points in the plane whose k -Gabriel graphs have diameter $\lceil \frac{n-1}{k+1} \rceil$.

Proof. The upper bound follows from Theorem 4 and Lemma 1. As for the second part of the statement, in the example of Theorem 4, $d_c(p_1, p_n) = \lceil \frac{n-1}{k+1} \rceil$. $\square$

4. Chromatic number

A j -coloring of a graph $G = (V, E)$ is a mapping $f: V \rightarrow \{1, 2, \dots, j\}$ such that $f(v) \neq f(w)$ for every edge (v, w) of G . The chromatic number of G , denoted by $\chi(G)$, is the minimum j such that G is j -colorable.

Theorem 5. For any set S of n points in the plane and $k \leq \lceil n/2 \rceil - 1$, $\chi(k\text{-GG}(S)) \leq \chi(k\text{-DG}(S)) \leq 6(k + 1)$.

Proof. The number of edges of $k\text{-DG}(S)$ does not exceed $3(k + 1)n - 3(k + 1)(k + 2)$ [1]. Consequently, the graph contains a vertex of degree at most $6k + 5$. Observe that, if (p_i, p_j) is an edge of $k\text{-DG}(S)$, this edge is also present in $k\text{-DG}(S \setminus \{p_l\})$ for any $p_l \in S$ ($p_l \neq p_i, p_j$). Thus, if $k\text{-DG}(S) \setminus S'$ is an induced subgraph of $k\text{-DG}(S)$ on n' vertices, then it is a subgraph of $k\text{-DG}(S \setminus S')$ and it has no more than $3(k + 1)n' - 3(k + 1)(k + 2)$ edges. Hence we can color $k\text{-DG}(S)$ with $6k + 6$ colors applying the minimum degree greedy algorithm [9]. $\square$

Next we describe a point set whose k -Gabriel graph has high chromatic number:

Proposition 1. For any $n \geq 3$ and $k \leq \frac{n-3}{3}$, there exist sets of n points in the plane whose k -Gabriel graphs have chromatic number at least $3k + 3$.

Proof. We use a construction described in [4] and illustrated in Fig. 3 (left). The points P, Q, R form an equilateral triangle. $\widehat{PQ}$ denotes the arc of the circle centered at R with endpoints P, Q , and $\widehat{QR}$ and $\widehat{RP}$ are defined analogously. The set S consists of the following $3k + 3$ points: a group of $k + 1$ points $p_1, p_2, \dots, p_{k+1}$ on $\widehat{PQ}$ and close to P , $k + 1$ points $q_1, q_2, \dots, q_{k+1}$ on $\widehat{QR}$ and close to Q , and $k + 1$ points $r_1, r_2, \dots, r_{k+1}$ on $\widehat{RP}$ and close to R . There is a way to choose the exact positions of these points so that, for every $x, y \in S$, the closed disk centered at the midpoint of x, y with both x and y on its boundary contains at most $k + 2$ points from S (see [4] for details). Thus, for every $x, y \in S$, the edge (x, y) belongs to $k\text{-GG}(S)$. Then $k\text{-GG}(S)$ is a clique and has chromatic number $3k + 3$.

We finally argue that the point set as described can be perturbed in a way that the resulting set of points is in general position and $k\text{-GG}$ remains the complete graph. Since in the original construction all edges belong to $k\text{-GG}(S)$, we only need to show that none of these edges disappear from the graph after perturbing S . This is indeed the case because the region associated to every edge (the diametral disk containing at most k additional points from S) is closed, so points that are outside the disk in the original construction remain outside the new disk if the points are infinitesimally perturbed. Additionally, notice that, if the given n is greater than $3k + 3$, it suffices to add to S additional points far from $p_1, p_2, \dots, p_{k+1}, q_1, q_2, \dots, q_{k+1}$, and $r_1, r_2, \dots, r_{k+1}$, so that the initial adjacencies are preserved. $\square$

For k -Delaunay graphs we provide a better construction:

Proposition 2. For any $n \geq 4$ and $k \leq \frac{n-4}{4}$, there exist sets of n points in the plane whose k -Delaunay graphs have chromatic number at least $4k + 4$.

Proof. Let S be the point set in Fig. 3 (right). The points P, Q, R , form an equilateral triangle, U is the midpoint of R and Q , and T is a point on the vertical line through R such that $|RT| = |RP|$. The set S consists of $k + 1$ points $p_1, p_2, \dots, p_{k+1}$

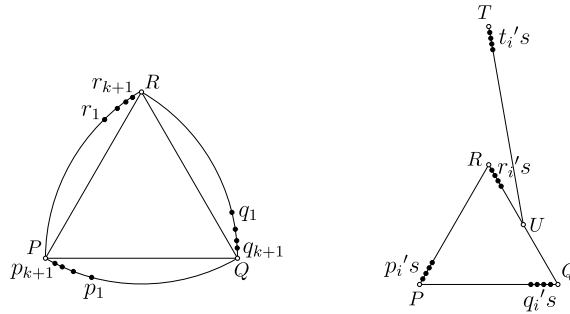


Fig. 3. Point sets of Propositions 1 and 2 for $k = 3$.

on the segment $\overline{RP}$ placed very close to P , $k + 1$ points $q_1, q_2, \dots, q_{k+1}$ on $\overline{PQ}$ very close to Q , $k + 1$ points $r_1, r_2, \dots, r_{k+1}$ on $\overline{QR}$ very close to R , and $k + 1$ points $t_1, t_2, \dots, t_{k+1}$ on $\overline{TU}$ very close to T .

This construction was originally described in [15], where the authors showed that, for every pair of points in S , there exists a circle through them containing at most k additional points of S . Consequently, $k\text{-DG}(S)$ is the complete graph and its chromatic number is $4k + 4$. Notice that the construction can be perturbed to avoid that three or more points are collinear. $\square$

5. Constrained geometric thickness

Let us recall that, given a geometric graph G , the *constrained geometric thickness* of G , denoted by $\theta_c(G)$, is defined as the minimum number of layers necessary to partition the edges of G so that no two edges of the same layer cross. Since $\text{GG}(S)$ and $\text{DT}(S)$ are plane, the constrained geometric thickness of these graphs is 1. In this section we give bounds on $\theta_c(k\text{-DG}(S))$ and $\theta_c(k\text{-GG}(S))$, for $k \geq 1$. We also tighten these bounds for the particular case $k = 1$. This extra effort is motivated by the fact that order-1 proximity graphs (together with order-0 proximity graphs) are the most commonly used in applications [1,13,20,21].

5.1. Constrained geometric thickness of $k\text{-DG}(S)$ and $k\text{-GG}(S)$

Let us introduce some definitions and recall some properties of $k\text{-DG}(S)$.

Edges of $\text{DT}(S)$ are said to have *order* 0. The edges of *order* $k \geq 1$ are those belonging to $k\text{-DG}(S)$, but not to $(k - 1)\text{-DG}(S)$.

Let (p_i, p_j) be an edge of order k . Then (p_i, p_j) is an edge in $\text{DT}(S \setminus \{p_{l,1}, p_{l,2}, \dots, p_{l,k}\})$ for some $\{p_{l,1}, \dots, p_{l,k}\} \in S$. We will say that (p_i, p_j) is *generated* by $\{p_{l,1}, \dots, p_{l,k}\}$. It holds that: (i) (p_i, p_j) is generated by $\{p_{l,1}, \dots, p_{l,k}\}$ if and only if there exists a circle through p_i and p_j whose interior contains $p_{l,1}, \dots, p_{l,k}$ and no other point in S ; (ii) if (p_i, p_j) is generated by $\{p_{l,1}, \dots, p_{l,k}\}$, then $(p_{l,v}, p_i)$ and $(p_{l,v}, p_j)$ are edges in $(k - 1)\text{-DG}(S)$ for all $v \in \{1, \dots, k\}$.

In order to prove Theorem 6 we use the following well-known geometric fact:

Observation 1. Let (p_i, p_j) and (p_l, p_m) be two crossing edges. Either every circle through p_i and p_j contains p_l or p_m , or every circle through p_l and p_m contains p_i or p_j .

Theorem 6. For any set S of n points in the plane and $k \leq \lceil n/2 \rceil - 1$, $\theta_c(k\text{-DG}(S)) \leq \frac{\chi^2((k-1)\text{-DG}(S)) + \chi((k-1)\text{-DG}(S))}{2}$.

Proof. Consider a $\chi((k - 1)\text{-DG}(S))$ -vertex coloring f of $(k - 1)\text{-DG}(S)$. We use f to construct a $\frac{\chi^2((k-1)\text{-DG}(S)) + \chi((k-1)\text{-DG}(S))}{2}$ -coloring of the edges of $k\text{-DG}(S)$ such that within each color class no two edges cross. This edge coloring is defined as follows: if (p_i, p_j) is an edge of $k\text{-DG}(S)$, the color assigned to (p_i, p_j) is the tuple $\{f(p_i), f(p_j)\}$.

Let us prove that no two edges of the same color cross. Suppose that (p_i, p_j) and (p_l, p_m) are two crossing edges in $k\text{-DG}(S)$, where (p_i, p_j) has order s and (p_l, p_m) has order t , with $0 \leq s, t \leq k$. By Observation 1, either $C(p_i, p_j)$ contains p_l or p_m , or $C(p_l, p_m)$ contains p_i or p_j . Without loss of generality, let us assume that the circle $C(p_i, p_j)$ contains p_l in its interior. Then p_l is connected to p_i and p_j in the graph $(s - 1)\text{-DG}(S) \subseteq (k - 1)\text{-DG}(S)$. Therefore $f(p_l) \neq f(p_i), f(p_j)$. $\square$

Corollary 5. For any set S of n points in the plane and $k \leq \lceil n/2 \rceil - 1$, $\theta_c(k\text{-GG}(S)) \leq \theta_c(k\text{-DG}(S)) \leq 18k^2 + 3k$.

We point out that the proof of Theorem 6 is constructive, that is, it describes a method to decompose $k\text{-DG}(S)$ and $k\text{-GG}(S)$ into at most $18k^2 + 3k$ plane layers. This decomposition might nevertheless be non-optimal.

In the case of the constrained geometric thickness our worst-case upper and lower bounds do not have the same order of magnitude:

Proposition 3. For any $n \geq 3$ and $k \leq \frac{n-3}{3}$, there exists a set S of n points in the plane such that $\theta_c(k\text{-DG}(S)) \geq \theta_c(k\text{-GG}(S)) \geq \frac{3k}{2}$.

Proof. Consider the point set in the proof of Proposition 1. We rename the points as $s_1, s_2, \dots, s_{3k+3}$ (this is also the counterclockwise sequence along the convex hull of the point set). The edges $(s_1, s_{\lfloor \frac{3k+5}{2} \rfloor}), (s_2, s_{\lfloor \frac{3k+7}{2} \rfloor}), \dots, (s_{\lfloor \frac{3k+3}{2} \rfloor}, s_{2\lfloor \frac{3k+3}{2} \rfloor})$ belong to the k -Gabriel graph and are pairwise crossing. Therefore the thickness of the graph is at least $\lfloor \frac{3k+3}{2} \rfloor$. $\square$

We tend to believe that our worst-case lower bounds on the constrained geometric thickness of the k -Delaunay and Gabriel graphs are closer to the true values than our upper bounds. The following proposition might give some evidence in this direction.

Proposition 4. Let S be a set of n points in the plane. The graph $k\text{-DG}(S)$ contains at most $2k + 1$ pairwise crossing edges.

Proof. Suppose that the graph $k\text{-DG}(S)$ contains l pairwise crossing edges. We consider the complete graph H having these edges as vertex set and oriented as follows: we orient an edge from (p_l, p_m) to (p_i, p_j) if $C(p_i, p_j)$ contains p_l or p_m . By Observation 1, every edge of H is oriented. Furthermore, if a vertex (p_i, p_j) of H has in-degree d , then $C(p_i, p_j)$ contains at least d points, since the d edges oriented to (p_i, p_j) are pairwise crossing, which implies that they do not have endpoints in common. Thus every vertex in H has in-degree at most k . Counting the edges of H , we have that $\binom{l}{2} \leq lk$. Consequently, $l \leq 2k + 1$. $\square$

5.2. Constrained geometric thickness of 1-DG(S) and 1-GG(S)

In this subsection we improve to 4 the upper bound $\theta_c(1\text{-DG}(S)) \leq 21$ given by Corollary 5. First we make some observations on the structure of 1-DG(S).

Let (p_i, p_j) be an edge of order 1. Then (p_i, p_j) is an edge in $\text{DT}(S \setminus p_l)$ for a certain $p_l \in S$. We will say that (p_i, p_j) is generated by p_l . Observe that: (i) (p_i, p_j) is generated by p_l if and only if there exists a circle through p_i and p_j whose interior contains p_l and no other point in S ; (ii) every edge of order 1 is generated by at most one point on each side of the line determined by the edge; (iii) if (p_i, p_j) is generated by p_l , then (p_l, p_i) and (p_l, p_j) are edges in $\text{DT}(S)$. (See [1].)

As observed in [3], Observation 1 implies that the crossings in 1-DG(S) have a certain structure:

Corollary 6. Let $(p_i, p_j), (p_l, p_m)$ be two crossing edges in 1-DG(S). If both edges have order 1, then one of them can only be generated by the endpoints of the other. If (p_l, p_m) has order 0 and (p_i, p_j) has order 1, then (p_i, p_j) can only be generated by p_l and p_m .

We now prove the main result of this subsection:

Theorem 7. For any set S of n points in the plane, $2 \leq \theta_c(1\text{-DG}(S)) \leq \chi(\text{DT}(S)) \leq 4$.

Proof. The graph $\text{DT}(S)$ is maximal planar, hence each edge of order 1 crosses at least one edge in $\text{DT}(S)$. Since the number of edges of order 1 is strictly greater than zero [1], at least two layers are needed.

We now prove the upper bound. Let f be a $\chi(\text{DT}(S))$ -coloring of the vertices of $\text{DT}(S)$. We define a $\chi(\text{DT}(S))$ -coloring of the edges of 1-DG(S) as follows. Let (p_i, p_j) be an edge of 1-DG(S). If (p_i, p_j) has order 1 and is generated by p_l , we assign it the color $f(p_l)$ (if (p_i, p_j) is generated by two points of distinct colors, we arbitrarily assign one of the two colors). If (p_i, p_j) belongs to $\text{DT}(S)$, we assign it an arbitrary color different from $f(p_i)$ and $f(p_j)$.

Next we prove that each color class is plane.

Suppose that (p_i, p_j) and (p_l, p_m) are two crossing edges of order 1. By Corollary 6, one of them can only be generated by the endpoints of the other. Let us assume that this is the case of edge (p_i, p_j) . Then (p_i, p_j) has color $f(p_l)$ or $f(p_m)$. Since the points generating (p_l, p_m) are connected to both p_l and p_m in $\text{DT}(S)$, their color is different from $f(p_l)$ and $f(p_m)$. Consequently, (p_l, p_m) is assigned a color different from $f(p_l)$ and $f(p_m)$.

Suppose that (p_i, p_j) and (p_l, p_m) are two crossing edges, where (p_i, p_j) has order 1 and (p_l, p_m) has order 0. The color of (p_l, p_m) is different from $f(p_l)$ and $f(p_m)$. By Corollary 6, (p_i, p_j) can only be generated by p_l and p_m . Hence its color is $f(p_l)$ or $f(p_m)$. $\square$

Corollary 7. For any set S of n points in the plane, $\theta_c(1\text{-GG}(S)) \leq \chi(\text{DT}(S))$.

We now give a worst-case lower bound on the constrained geometric thickness of 1-DG(S) and 1-GG(S):

Proposition 5. For any $n \geq 6$, there exists a set S of n points in the plane such that $\theta_c(1\text{-DG}(S)) \geq \theta_c(1\text{-GG}(S)) \geq 3$.

Proof. Fig. 4 shows a set of 6 points whose 1-Gabriel graph contains three pairwise intersecting edges. Thus its constrained geometric thickness is at least three. For larger values of n it suffices to add $n - 6$ points outside the disks. $\square$

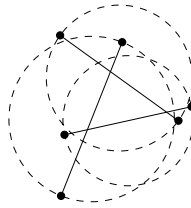


Fig. 4. Example of a set of 6 points whose 1-Gabriel graph has constrained geometric thickness at least 3.

6. Final remarks

We have studied several properties of two fundamental higher order proximity graphs.

As for open problems, a natural one is to close the gaps between the lower and upper bounds on the spanning ratio of k -Gabriel graphs and on the constrained geometric thickness of k -Gabriel and k -Delaunay graphs. In both cases we are inclined to think that the lower bounds are closer to the true values.

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